SOKOINE UNIVERSITY OF AGRICULTURE

School of Engineering and Technology / College of Natural and Applied Sciences

Department of Informatics and Information Technology (DIIT)

BIT 111: MICROCOMPUTERS AND APPLICATIONS

TAKEHOME HOMEWORK ASSIGNMENT 1 (TOPIC 1)

Theme: INTRODUCTION TO COMPUTERS, GENERATIONS, PIONEERS & CLASSIFICATION

Course Instructor: Madam Bertha Msuliche Lebalwa (MSc)
Curriculum Coverage: Subtopic 1.1 (Computer Generations), Subtopic 1.2 (History & Pioneers), Subtopic 1.3 (Classifications)
Assessment Weight: Coursework Component (Official Scheme: Assignments 5%, Tests 15%, Practical Report 10%, Practical Test 30%, UE 40% - subject to instructor review)
Submission Deadline: End of Week 2 by 17:00 EAT | Total Allocation: 100 Marks

INSTRUCTIONS TO CANDIDATES:
1. This assignment must be solved individually. Plagiarism, peer copying, or unreferenced AI generation will result in an automatic score of zero and disciplinary reporting under SUA Examination Regulations.
2. Answer ALL questions in Sections A, B, and C. Allocate your effort according to the mark allocations shown in brackets.
3. Show all intermediate formulas, binary conversions, architectural diagrams, command syntax, and arithmetic calculations clearly.
4. Where design questions require software configurations, provide exact command line syntax (Bash/DOS) or functional formulas (Excel) as specified.
5. Submit your completed work as a clean, stapled technical report or upload the single consolidated PDF to the SUA Virtual Learning Environment (Moodle) before the deadline.

SECTION A: FOUNDATIONAL & HISTORICAL CONCEPTS [25 MARKS]

Answer all five questions in this section. Concise, technically accurate answers are required.

Question 1 [5 Marks]: Differentiate clearly between data and information within an agricultural enterprise context. Illustrate your answer by outlining the four stages of the Information Processing Cycle (IPOS) as applied to an automated grain moisture monitoring station at SUA Mazimbu Farm.

Question 2 [5 Marks]: Compare First-Generation and Second-Generation computers across the following specific dimensions: (a) Primary active switching electronics, (b) Primary internal memory technology, (c) Typical physical power and cooling requirements, and (d) Primary programming paradigm. Provide one canonical machine example for each generation.

Question 3 [5 Marks]: Analyze the historical contribution of Charles Babbage and Augusta Ada Lovelace. Why is the Analytical Engine universally recognized as the theoretical prototype of modern general-purpose digital computers, and what conceptual breakthrough did Lovelace contribute regarding algorithmic execution?

Question 4 [5 Marks]: Explain the fundamental architecture proposed by John von Neumann in his 1945 EDVAC report. What is the 'stored-program concept', and what physical phenomenon constitutes the 'Von Neumann Bottleneck' in contemporary microcomputer processing?

Question 5 [5 Marks]: Distinguish between Analog, Digital, and Hybrid computing systems based on signal representation and mathematical operational behavior. Provide an example where a hybrid computer is strictly superior to a purely digital computer in an agricultural engineering application.

SECTION B: QUANTITATIVE & PERFORMANCE ANALYSIS [35 MARKS]

Question 6 [10 Marks]: A microcomputer workstation at the SUA Remote Sensing Laboratory possesses a multi-core processor operating at a base clock frequency of 3.60 GHz. An agricultural image classification algorithm requires the execution of 4.5×10^{10} machine instructions.
(a) If the average Cycles Per Instruction (CPI) for this benchmark program is 1.25, calculate the total CPU execution time in seconds. [4 Marks]
(b) Compute the effective MIPS (Million Instructions Per Second) rating of the processor running this specific workload. [3 Marks]
(c) If architectural compiler optimizations reduce the required instruction count to 3.2×10^{10} while increasing the CPI slightly to 1.30 due to cache branch misses, calculate the percentage speedup achieved. [3 Marks]

Question 7 [10 Marks]: The Department of Informatics and Information Technology is profiling a crop yield simulation code. Currently, 75% of the total execution time is spent in a matrix multiplication routine that can be parallelized across multi-core processors, while the remaining 25% consists of strictly sequential I/O operations.
(a) Using Amdahl's Law, derive and compute the theoretical maximum speedup achievable if the matrix routine is executed across an 8-core CPU. [4 Marks]
(b) Calculate the absolute asymptotic maximum speedup possible on an infinite number of processor cores ($N \to \infty$). [3 Marks]
(c) Explain two physical overhead factors that prevent real-world multi-core microcomputers from reaching the theoretical Amdahl speedup. [3 Marks]

Question 8 [8 Marks]: Analyze the evolution of semiconductor scaling as described by Moore's Law and Dennard Scaling.
(a) Formulate Gordon Moore's empirical 1965/1975 observation regarding transistor density and economic cost per transistor. [3 Marks]
(b) Explain why processor manufacturers shifted from aggressive clock frequency scaling (which peaked around 3.8–4.0 GHz in the mid-2000s) to multi-core architectures. Reference the 'Power Wall' formula $P = C \cdot V^2 \cdot f$ in your explanation. [5 Marks]

Question 9 [7 Marks]: Perform the following technical data representation and bus capacity calculations:
(a) A 32-bit address bus can uniquely address a maximum physical memory space. Calculate this theoretical limit in exact Gibibytes (GiB). [3 Marks]
(b) A modern 64-bit microcomputer system implements 48 physical address lines. Calculate the maximum physical RAM addressable by the CPU hardware in Terabytes (TB) and Tebibytes (TiB). [4 Marks]

SECTION C: APPLIED ENTERPRISE & INFRASTRUCTURE DESIGN [40 MARKS]

Question 10 [15 Marks]: The SUA School of Engineering and Technology intends to procure new computer infrastructure for two distinct facilities: (Facility 1) A Computer-Aided Design (CAD) and drone aerial photogrammetry laboratory for precision farming; (Facility 2) An automated field weather telemetry network spanning 12 outlying experimental agricultural stations in Morogoro Rural.
(a) Specify and justify the optimal computer classification (e.g., Workstation vs Desktop PC vs Single-Board Computer vs Embedded Microcontroller) for each facility. [5 Marks]
(b) Produce a comparative specification matrix evaluating both choices across: Processor class, memory capacity and ECC requirements, graphics accelerator, thermal/environmental tolerance, power source, and operating system. [6 Marks]
(c) Outline the network role (Client, Peer, or Dedicated Server) each machine plays in the overall enterprise data collection pipeline. [4 Marks]

Question 11 [15 Marks]: Contrast the architectural paradigms of Supercomputers, Enterprise Mainframes, and Microcomputer Clusters.
(a) Discuss why a commercial banking transaction processing system (such as CRDB or NMB in Tanzania) relies on IBM z16 Mainframes with 99.999% ('five nines') availability rather than high-performance supercomputers. [6 Marks]

(b) Contrast the concepts of Reliability, Availability, and Serviceability (RAS) with Peak Floating-Point Operations Per Second (FLOPS). [5 Marks]
(c) Explain how a cluster of commodity microcomputers (Beowulf cluster) running Linux can economically perform agricultural genomics and climate forecasting previously restricted to monolithic supercomputers. [4 Marks]

Question 12 [10 Marks]: Terminal Diagnostics & Operating Environment Analysis:
A systems technician at SUA is auditing a microcomputer laboratory running Ubuntu Linux.
(a) Provide the exact terminal commands required to display: (i) Detailed CPU architecture and cache topology, (ii) Total and available RAM in human-readable megabytes/gigabytes, (iii) Motherboard BIOS/firmware vendor and version, and (iv) Storage block device hierarchy. [5 Marks]
(b) Interpret the hypothetical output from `lscpu` showing 'Model name: Intel Core i7-12700K', 'Thread(s) per core: 2', 'Core(s) per socket: 12', 'Socket(s): 1'. Calculate the total number of logical processors (threads) exposed to the operating system scheduler. [5 Marks]